

Aditya Subramanian

Junior Undergraduate
Department of Aerospace Engineering

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| in adityasubramanian

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2020 - Present	B.Tech	Indian Institute of Technology, Kanpur	8.21/10.00
2020	CBSE(XII)	PSBB LLA, Bangalore	96.4%
2018	CBSE(X)	PSBB LLA, Bangalore	95.6%

Scholastic Achievements

- Represented **IIT-Kanpur** at the fifth edition of **National Aerospace Conceptual Design Competition** in 2022
- Secured **AIR 2703** in **JEE-Advanced 2020** amongst 2.24 Lakh candidates, and **AIR 8001** in **JEE-Main 2020**
- Among **top 29** from **Bangalore** who qualified for nationals of **Indian Computing Olympiad** - INOI conducted in 2020
- One of **top 35** selected to attend the six-week **research internship**, Research Science Initiative - Chennai, **IIT Madras**, 2019

Experience

Research Intern, SURGE IIT Kanpur || Mentors: Prof Sanjay Mittal, Prof. Durgesh Vikram ☞ (May'22-Present)

Objective	- To build a computational model to predict the effect of road geometry on traffic, using equations of fluid dynamics
Approach	- Performed literature survey on the existing finite element formulation used to model traffic as fluid-like continuum - Ideated the modelling of a curve and of converging/diverging road patterns, and formalized possible solutions - Conducting numerical experiments on the developing model to verify and subsequently improve it

InterIIT Tech Meet 10.0



Gold Medal, Blue Yonder's Sustainable Supply Chain Case ☞ (Feb'22-March'22)

Objective	- Layout an extensive, time-phased 5-year roadmap with initiatives to achieve sustainable supply chains
Strategy	- Followed MAQ framework to suggest initiatives for the Indian transport sector to reduce its carbon footprint - Ideated intermodal shift, reducing 14.13% CO₂ emissions, and package return policy saving 10.4 Mn-Kg carbon - Proposed unified 3PL model and innovative solutions in the Warehousing, Transport, and Packaging sectors
Result	- Member of the team coming first amongst 20+ participating IITs in the 10th Inter-IIT Tech Meet

Key Projects

Summer Project - Game of Blocks, Programming Club, IIT Kanpur ☞ (June'21-July'21)

Objective	- To study and implement the concept of Blockchain and smart contract development with a flavor of Game Theory
Strategy	- Understanding various consensus mechanisms, learning Solidity and implementation of Smart Contracts on it - Introduction to Game theory and Mechanism Design and its applications in real world situations - Implementation smart contracts for Voting and Auction mechanisms on Solidity using concepts from game theory
Result	- Built a Smart-Contract-Hub on Github containing 5 auction, 12 voting mechanisms, and 3 school choice systems

Positions of Responsibility



• **Head, Public Relations, Techkriti '23** (July'22-Present)

Leadership	- Building a 3-tier team of 50+ members to organize Techkriti '23 with an forecasted budget of over ₹1.5 Crore - Coordinating with a core team with heads from 9 administrative portfolios to organize and conduct the festival
Initiatives	- Renovating embassy deals to bring in enhanced international collaborations and approaching 200+ embassies - Scouting and inviting accomplished individuals from across the globe like Nobel laureates for giving 'Techtalks' - Planning to collaborate with start-ups, companies and universities to bring in cutting edge technical exhibitions

• **Secretary, Media and Publicity, Games and Sports Council** (March'21-Apr'22)

Initiatives	- Managed all social media handles of the council, followed by more than 13,500 unique accounts - Standardized using professional tools like Business Suite to organize, manage and execute publicity campaigns - Organized and conducted fortnightly events including talks, workshops and inter and intra-hall competitions
Impact	- Increased the Instagram reach by ~900 accounts (81% YoY) and revived the council's dormant YouTube channel

Technical Skills

- **Software and Libraries:** Canva, Business Suite, Git, Github, ROS, Selenium, Gazebo, Micro-Cap, LabVIEW
- **Programming Languages:** C, C++ (20), Python, L^AT_EX, Solidity, Julia, HTML, JavaScript, SQL

Extra-Curricular Activities

Social work	- UG Nominee to Women's Cell (2022), working to mitigate sexual harassment & gender-based discrimination
Technical	- Mentoring Deep Dive into Web3.0 project and contributed to Github repositories of Programming Club as a secretary - Worked on open source solutions for UAVs as a technical team member of the institute's Aerial Robotics Team
Cultural	- Participated in the Dance and Drama showcases for Freshers' and won 2nd Prize in Pun Battle event - Contingent member to the Inter IIT Scrabble League 2021, team placed overall fourth amongst competing IITs
Managerial	- Head Prefect , leading the student council of 30 members, PSBB LLA Bangalore from 2019-2020 - Helped in the management of guests for 'Techtalks', Techkriti '21 and anchored for Comedy night, Techkriti '22